

State of the Art: Economic Development Through the Lens of Paintings

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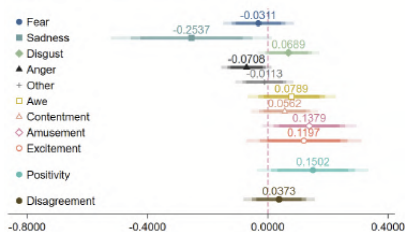
Research Question



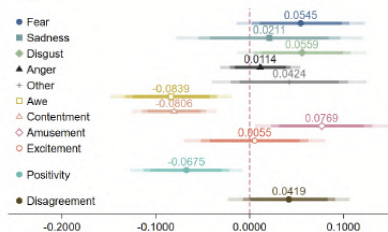
Figure: Urban Prosperity vs. Rural Poverty as Examples of the Illustrative Properties of Artwork

- Do the emotions conveyed in visual artworks reflect socioeconomic context?
 - Can artwork be used as a historically accurate, high-frequency measure of economic and societal conditions?

Results



(a) GDP per capita

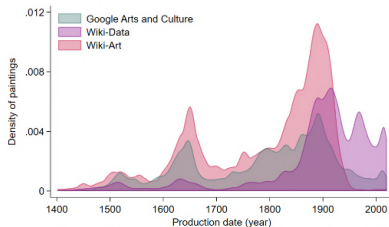


(b) Uncertainty

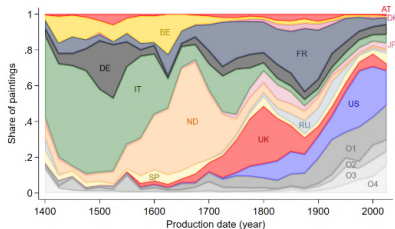
Figure: Excerpt from the regression results. The bands indicate different levels of confidence intervals (10%, 5%, 1%). Results highlight how different economic and societal shocks are reflected intuitively in changes to emotions.

- Fixed-effects regression using economic variables to explain changes in nine aggregated emotion probability scores as outcomes (**contentment**, **excitement**, **amusement**, **awe**, **fear**, **anger**, **sadness**, **disgust**)

Overall Sample



(a) Time of production



(b) Selected countries

- 630k paintings from open-access repositories (Google Arts & Culture, WikiArt, Wikidata)
 - 29k artists
 - Time frame: 1400–2000
 - 34 countries with >1k paintings
- Physical features, genre, artistic movement (post-impressionism, mannerism)

Annotated Subset

- 80k annotated paintings
- 15–25 annotations per painting choosing one of 9 emotions and a free-text field
 - Aggregation
 - Probabilistic emotion vectors for each image, using the distribution of emotion annotations across annotators
 - For each free-text annotation: conversion into a probabilistic emotion vector using an LLM

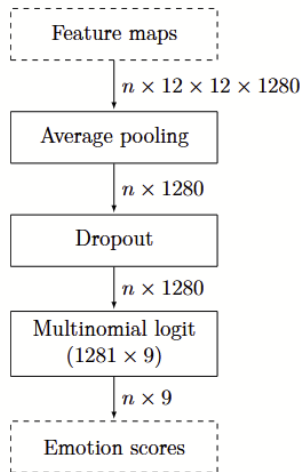
CNN

- Model

- Convolutional EfficientNet architecture pre-trained on ImageNet (output layer removed)
 - Learns generic features (textures, motifs, shapes) employed by artists

- Training

- First, random initialization of output layer with frozen encoder layers to stabilize parameters
- Second, fine-tuning of CNN encoder by unfreezing weights and reducing learning rate to transfer from (original) photo to artwork domain



(b) Regression head

Relation to Class Materials

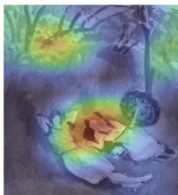
- CNNs are not 100% state-of-the-art anymore with the emergence of transformer architectures
 - However, publishing and researching takes time, introducing a time-lag between cutting-edge methods and applied papers
 - CNNs are well-researched, especially regarding explainability
- Application of GTM
 - Outcome
 - Which words are associated with which painting, i.e. describe the effect the painting has on the viewer
 - Annotated dataset as training data and probabilistic vector of emotions as semi-supervised inputs for $k=9$, emotions as the latent topics
 - (Subset of) metadata annotations (years, artists, countries, etc.) as latent, prediction, and outcome covariates
 - Free-text embeddings or bag-of-words outputs as response variables

Summarized Evaluation

- Idea is well-established
- Execution relies on advanced modeling technique at large scale
- Results seem highly significant and intuitive
- Most impressive is validation
 - [Assumptions](#)
 - [Statistical checks](#)
 - CNN behavior

CNN Behaviour

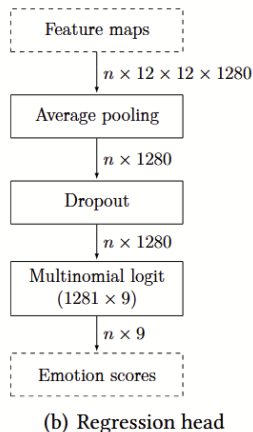
- Over- vs. underfitting (70/15/15 split)
 - Train vs. test KL-divergence distributions align overall, across time, and across genres (no strong overfitting)
- CAM
 - Class Activation Mapping highlights regions most relevant to the final emotion prediction, enabling visual assessment of whether filters behave as expected, which they appear to.



(a) Contentment



(e) Fear



Broader Implications

- Research: computational analysis of even very abstract art works well
- Research & Society: confirms art as a reflection of broader society; potential as an accurate high-frequency measure of developments
 - However, art remains deeply personal
 - Therefore, the approach is mostly historical and very large scale—tendencies mitigate this, where the paper is somewhat overoptimistic
 - Sources designed for this purpose are more suitable (e.g., newspapers)
- Policy implications: art is important, and measures should be taken to promote its creation

Questions for you

- What would be a potential prediction outcome in the proposed GTM model?
- Do you think the assumption: "contemporary emotions = emotions of initial audience / goal of artist" is valid?

Validation: Statistical Checks

- Error structure: secondary emotions are harder to capture; neutral emotions are overpredicted relative to distinctive ones
- Sample bias checks:
 - Preservation bias toward famous artists (selling records show $\sim 10\%$ premium)
 - Limited differences in emotional/technical approach of such artists; overall sample deemed “good”
- Fixed-effects, clustered error model
 - Controls for confounders at the artist, artistic movement, country, and year levels

Assumptions & Annotations

- Artists translate societal and economic conditions into visual form through their paintings.
- The paper inverts the process by inferring prevailing societal and economic circumstances from the emotions expressed in the artworks.
- Therefore assumptions:
 - Sample of art is representative
 - Validation: Sample size is very large and comparing artwork in sample with data from transactions
 - Contemporary audience emotions = emotions of initial audience / goal of artist
 - Validation: Vector of emotions from annotators' free-text vs. vector of emotions conveyed by artists' own descriptions, both annotated by LLMs (available for a small subset of artists)
 - To assess paintings using a CNN, a systematic formalism must underlie artistic expression and be learnable by the algorithm
 - Validation: Academic literature on art decomposes art into Principles of Art and Elements of Art, which associate certain techniques with corresponding emotional effects

Annotation & Augmentation

- Consolidating genres, movements, etc.
 - Use of NER-algorithms to find artists' names, match with structured Wikipedia information
- Detecting duplicates
 - Use of 1260-dimensional CNN outputs to compare paintings and filter out based on cosine similarity